Probability, conditional probability and Bayes' Theorem

* Probability

Probability measures the likelihood of an event occurring.

Its value ranges from 0 to 1.

- P(A) = 0: Impossible event.
- P(A) = 1: Certain event.

Basic formula:

P(A) = Number of favorable outcomes / Total Number of Equally likely outcomes

Example: When rolling a die, the probability of getting an even number is:

P(A) = 3/6 = 1/2

* Rules of Probability

A. Complement Rule

$$P(A^c) = 1 - P(A)$$

B. Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

For mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

C. Multiplication Rule

$$P(A \cap B) = P(A) \times P(B|A)$$

For independent events:

$$P(A \cap B) = P(A) \times P(B)$$

* Conditional probability

conditional probability is the probability an event occurring given that another event has already occured.

Formula:

$$P(A|B) = P(A \cap B) / P(B)$$

where $P(B) > 0$.

* Independent Events

Two events are independent if the occurence of one does not change the probability of the other.

formula:

$$P(A \cap B) = P(A) \times P(B)$$

Example: A fair coin is tossed twice.

P(Head on both tosses) = 1/2 x 1/2 = 1/4

* BAYES' THEOREM

Bayes' theorem calculates the probability of an event using prior information and new evidence.

formula:

$$P(A|B) = [P(B|A) \times P(A)] / P(B)$$

where:

- $P(A|B)$: Posteria probability.
- $P(A|B)$: Likelihood.
- $P(A)$: prior probability.
- $P(B)$: probability of the evidence.

Applications: Medical diagnosis, machine learning, spam detection, and decision-making.

Conclusion:

statistics helps us collect, organize, and analyze data, while probability helps us measures uncertainty and calculate the likelihood of events. Both are important for data analysis and decision-making.